

equation used to calculate the temperature from ADC value is:

$$\text{temp} = \left[\left\{ 5.0 \times \text{analogRead}(\text{TemperaturePin}) \right\} / 1024 \right] \times 100.0$$

LCD. A 16x2 liquid crystal display (LCD1) is used to display the temperature. Various messages displayed on LCD1 are given in the testing section.

Buzzer. The piezoelectric buzzer produces sound based on reverse piezoelectric effect. Generation of pressure variation or strain by the application of electric potential across a piezoelectric material is the underlying principle.

The buzzer can be used to alert a user of an event corresponding to a switching action, counter signal or sensor input. It consists of piezo crystals between two conductors. When a potential is applied across these crystals, these push on one conductor and pull on the other. This push-and-pull action results in a sound wave.

Arduino Uno. Arduino Uno is a microcontroller (MCU) technology based on ATmega328 series. It contains 14 digital input/output pins (of

which six can be used as PWM outputs), six analogue inputs, one 16MHz crystal oscillator, USB connection, power jack, ICSP header and reset button. (Arduino Uno R3 was used during testing.)

Preset. A preset is a three-terminal variable resistor with a rotating contact that makes an adjustable voltage divider. If only two terminals are utilised, one end and the wiper, it behaves like a variable resistor or rheostat. Here, it is used to control the contrast of LCD1.

Software. Programming is done on Arduino version 1.8.6 integrated development environment (IDE). Analogue pins A0 to A5 on Arduino Uno are analogue input pins. After running the code, you will see voltage values on the serial monitor of the PC.

LM35 sensor is connected to pin A0, and the buzzer is connected to pin 9 of Arduino Uno, respectively.

Construction and testing

Make connections as per the given circuit diagram. Place the sensor near the

EFY Note
The source code of this project is available for free download at source.efymag.com

device to be monitored. Connect 5V USB power supply to Arduino board. Alternatively, you can use a 9V battery at V_{in} pin of Arduino Uno. Recorded temperature level will be displayed on LCD1.

Whenever there is a low temperature, below 10°C, 'Low' will be displayed on LCD1 along with a buzzer sound. If temperature is between 10°C and 40°C, 'Normal' will be displayed. If temperature is between 40°C and 100°C or above, 'High' or 'Very high' will be displayed on LCD1. **EFY**



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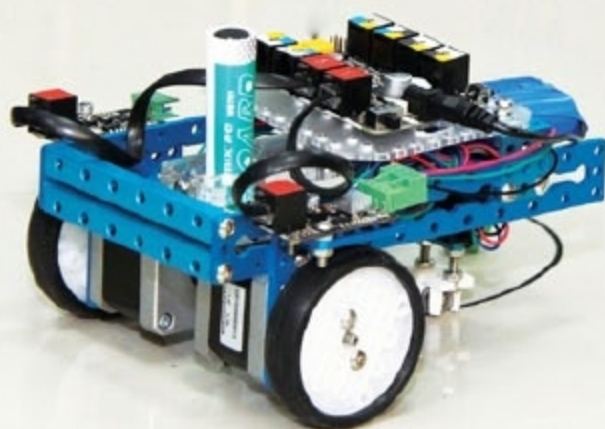
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